# Binary addition and overflow

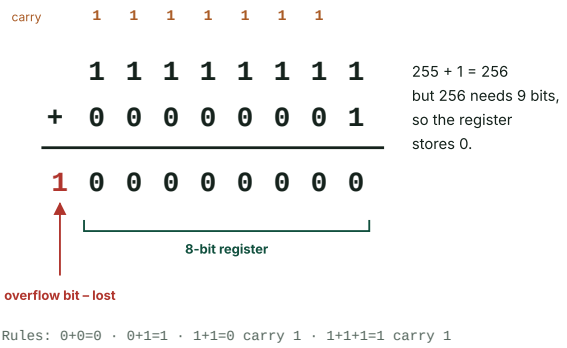
D05 FACT FILE

**Specification: students should be able to**

- Perform the addition of two bytes and explain the meaning of overflow

CCEA fact file: Digital Data

**Key words:** binary addition, carry, overflow, register, 8-bit




Adding 11111111 and 00000001 in 8 bits causes overflow

## Binary addition

Add from the right-hand column. 0+0 = 0, 0+1 = 1, 1+1 = 0 carry 1, and 1+1+1 = 1 carry 1. Write the carries clearly and check by converting the two numbers and the answer to denary.

## Overflow

Overflow happens when the result of a calculation needs more bits than are available. In an 8-bit register the largest value is 255, so any total above 255 produces a carry out of the 128 column that cannot be stored. The extra bit is lost and the stored result is wrong (e.g. 255 + 1 is stored as 0). This can make a program give incorrect results or fail.

## Key terms from the mark schemes

**Carry**
The digit passed to the next column when a binary column total exceeds 1.
Credited: U4 2025 Q4

**Overflow**
When a result is too large for the available bits, so the extra bit is lost and the result is wrong.
Credited: U4 2019 Q8, 2022 Q6, 2024 Q4, 2025 Q4

## Exam focus

Binary addition with overflow appears every year: 2019 Q8(d), 2022 Q6(c), 2023 Q7(d), 2024 Q4(d), 2025 Q4(b). Typical marks: 3 for the addition with carries and 1–2 for explaining overflow and its effect.

| | | |
|---|---|---|
| Unit 4 2025 Q4 | [15] | Binary conversion, addition and overflow, units, Unicode vs ASCII |
| Unit 4 2024 Q4 | [19] | Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables |
| Unit 4 2023 Q7 | [13] | Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow |
| Unit 4 2022 Q6 | [14] | Binary digit, 54 to binary, nibble, hex, overflow, storage capacity |
| Unit 4 2019 Q8 | [18] | Binary, hexadecimal and its uses, binary addition and overflow, logic gates |

## ? Check yourself

1. Add the binary numbers 00110101 and 00011010. Show the carries. [3]
2. Add 10110110 and 01100101. Show your working. [3]
3. Explain the meaning of overflow. [2]

**Answers:** 1. 00110101
+ 00011010
= 01001111
Carries shown into the 32 and 64 columns (1) / right-hand bits 1111 correct (1) / final answer 01001111 (1). Check: 53 + 26 = 79. 2. 10110110
+ 01100101
= 1 00011011
Carries shown (carries out of the 4, 32, 64 and 128 columns) (1) / 8-bit result 00011011 (1) / the carry out of the 128 column is a 9th bit – overflow has occurred (1). Check: 182 + 101 = 283, which is > 255; the stored 8 bits are 27. Accept the 9-bit answer 100011011 with overflow identified. 3. Overflow occurs when the result of a calculation is too large (1) / to be stored in the number of bits available, e.g. greater than 255 in an 8-bit register (1)